Problem Statement

TEAM NAME

Erdős

TEAM MEMBERS

Arpit Kumar
Roll No. 12340350

Contribution:
- RAG module
- POT + Code-sandbox module

Saurav Gupta
Roll No. 12341940

Contribution:
- Agentic correction module
- Learned verifier

THE PAPER	mmJEE-Eval: a test set of 1,460 JEE Advanced questions (2019–2025) in English and Hindi, given as images, covering Physics, Chemistry and Maths. It checks how well vision-language models solve complex reasoning problems.
THE PROBLEM	Free, open VLMs solve only 37–45% of the questions, while the best frontier models solve 77–84%. Additionally, VLMs perform worse on numerical questions—not because of flawed reasoning, but due to basic arithmetic calculation errors. Furthermore, while VLMs often recognize that their own answer is wrong, they manage to fix only 1–5% of these mistakes (the Metacognition Gap).
OUR GOAL	Design an architecture that helps open VLMs solve more questions without retraining them, and test it fairly on the unseen 2025 paper.

NOVELTY

01 The model writes Python code for every calculation and run it, so the arithmetic is exact.

02 A second model (Critic) checks each solution and points to the first wrong step; the solver then redoes it, up to 3 rounds.

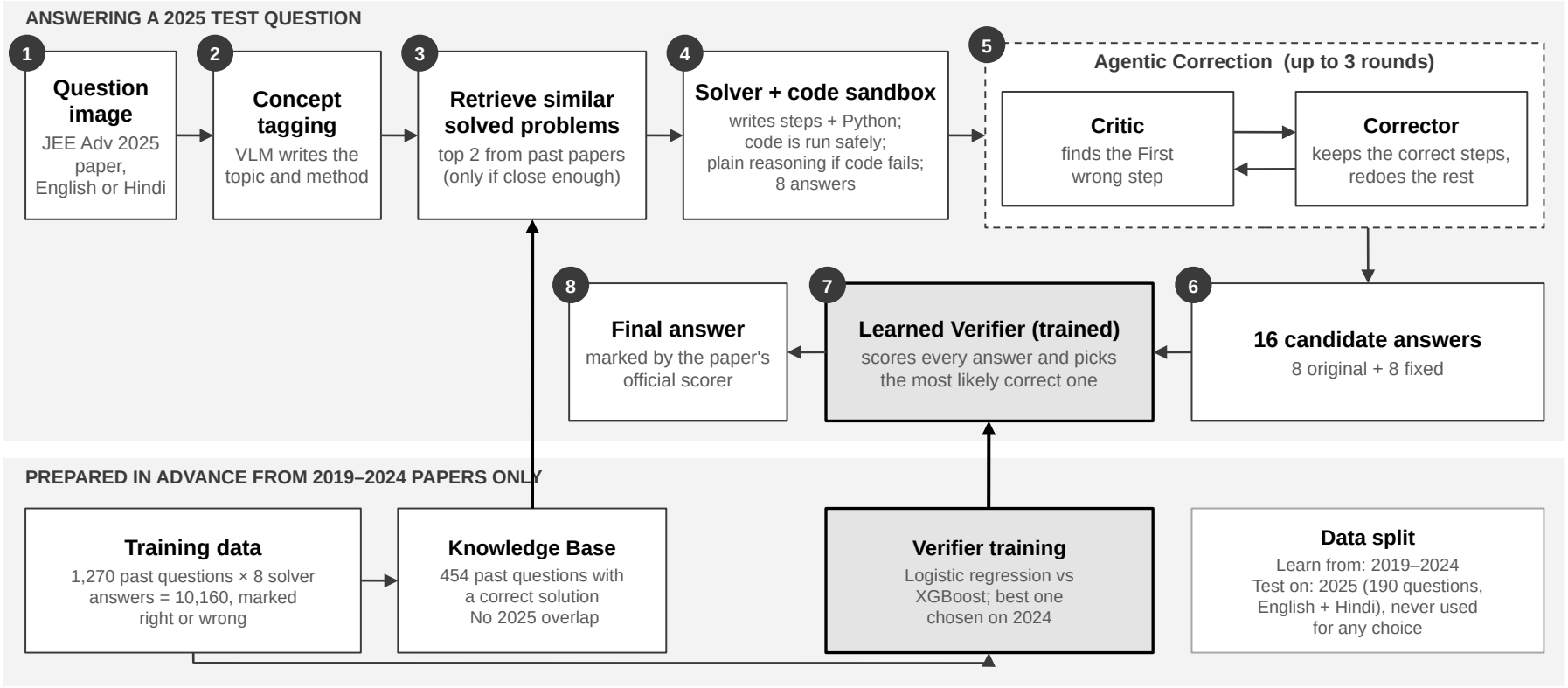
03 A trained classifier scores the 16 candidate answers and picks the best one.

04 Solved past questions that need a similar method are shown as examples.

05 All parts are combined into one end-to-end architecture: find similar examples → solve with code → check and fix → pick the best answer.

System Architecture

Frozen 4-bit VLMs on one RTX 3060 (12 GB): Qwen3-VL-8B (solver) · InternVL3.5-8B (critic)



Results and Findings (JEE Advanced 2025, 190 test questions)

ACCURACY (%) AS EACH PART IS ADDED

Setup	All	Numerical	Tokens
Baseline (one CoT answer)	43.2	40.2	2.7k
Majority vote of 20 answers	53.2	51.2	54.2k
+ Code sandbox	48.2	50.4	3.5k
+ Agentic correction	51.3	53.0	6.4k
+ Learned verifier	63.2	66.7	51.5k
+ RAG = Full framework	**65.3**	**69.0**	**54.1k**

- Model: Qwen3-VL-8B (4-bit AWQ, frozen, not retrained); critic: InternVL3.5-8B
- Hardware: one NVIDIA RTX 3060 (12 GB), vLLM 0.30
- Sampling: temperature 0.7, up to 4,096 tokens per answer; 8 solver answers per question
- Verifier training: 10,160 answers on 1,270 questions (2019–2024); best model chosen on 2024
- Test: 190 questions (95 English + 95 Hindi), scored with the paper's own scorer

KEY FINDINGS

- The full system solves 65.3% of questions, up from 43.2% with a single answer, and 12.1 points more than majority voting with the same compute ($p = 0.001$).
- Our setup is reliable: the paper's Qwen2.5-VL-7B scores 13.8% with our code, close to the 12.4% the paper reports.
- Numerical questions accuracy improved from 40.2% to 50.4%.
- The agentic correction loop fixes 14–19% of the mistakes it finds; the paper reports 1–5%.
- Picking the best of 16 answers (Learned verifier) gives the biggest jump (+11.8 points)
- Similar solved problems help only a little (+2.1 points): just 30 of 190 questions find a close match.
- No sign that the model memorised old papers: 43.9% on 2019–2024 vs 42.6% on 2025.
- Still hard: long answers cut off at 4,096 tokens, and Hindi and diagram questions.